

MATH-1014 T10B: EXERCISE

Advanced Convergence Tests and Strategies

April 27, 2026

Part A: Alternating Series & Absolute vs. Conditional Convergence

1. Determine whether the following series is absolutely convergent, conditionally convergent, or divergent:

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2 + 4}$$

2. Test for absolute and conditional convergence:

$$\sum_{n=1}^{\infty} (-1)^n (\sqrt{n+1} - \sqrt{n})$$

3. Determine whether the following series is absolutely convergent, conditionally convergent, or divergent:

$$\sum_{n=1}^{\infty} (-1)^n \left[e - \left(1 + \frac{1}{n} \right)^n \right]$$

4. Determine whether the following series converges or diverges:

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n} + (-1)^n}$$

5. Determine if the following alternating/oscillating series is absolutely convergent:

$$\sum_{n=1}^{\infty} |e^{-2n} \sin n|$$

6. Determine whether the following series converges or diverges, and provide justification:

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$$

Part B: The Ratio and Root Tests

7. Test the following series for absolute convergence:

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}$$

8. Test the convergence of the series:

$$\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$$

9. Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}$$

10. Use the Root Test to determine the convergence of the series:

$$\sum_{n=1}^{\infty} \left(\frac{n}{2n+1} \right)^n$$

11. Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \frac{2^n 5^n n!}{n^n}$$

Part C: Comparison Tests & Asymptotic Analysis

12. Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \frac{\sqrt{n^3 + 1}}{3n^3 + 4n^2 + 2}$$

13. Determine whether the following series converges or diverges:

$$\sum_{n=2}^{\infty} \frac{1}{(\ln n)^{\ln n}}$$

14. Determine whether the following series converges or diverges (Hint: Taylor/Maclaurin Expansion):

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \sin \left(\frac{1}{n} \right) \right)$$

15. Use an appropriate convergence test (e.g., Limit Comparison Test) to test the series:

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 1}$$

16. Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \frac{\ln n}{2n + 3}$$

17. Identify which of the following infinite series are convergent:

(i) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^5}}$ (ii) $\sum_{n=1}^{\infty} \frac{3^n n!}{(2n)!}$ (iii) $\sum_{n=1}^{\infty} \frac{\sin n}{n^2 + e^{-n}}$ (iv) $\sum_{n=1}^{\infty} \frac{2}{(2 + \frac{1}{n})^{2n}}$

Part D: The Integral Test & Improper Integrals

18. Determine whether the following series converges or diverges:

$$\sum_{n=3}^{\infty} \frac{1}{n \ln(n) \sqrt{\ln(\ln(n))}}$$

19. Find all values of $p \in \mathbb{R}$ for which the following series converges:

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$$

20. Determine whether the following improper integral converges or diverges:

$$\int_0^{\infty} \frac{e^{-x}}{\sqrt{x}} dx$$

21. Evaluate the following improper integral exactly, or show that it diverges:

$$\int_1^{\infty} \frac{\ln x}{x^2} dx$$

22. Evaluate the improper integral:

$$\int_2^4 \frac{3}{x\sqrt{x^2-4}} dx$$

23. Evaluate the improper integral:

$$\int_2^{\infty} \frac{5}{x^2\sqrt{x^2+5}} dx$$

24. Evaluate the improper integrals: (a) $\int_0^\infty \frac{1}{(x^2+1)^2} dx$ (b) $\int_0^2 \frac{4x}{\sqrt{4-x^2}} dx$

25. Determine the convergence or evaluate: $\int_0^{\pi/2} \frac{5 \sin x}{2 \cos x} dx$

26. Determine the constant C such that the following improper integral converges:

$$\int_0^\infty \left(\frac{2x}{x^2+2} - \frac{C}{x+3} \right) dx$$

27. Determine the constant k such that the following improper integral converges:

$$\int_0^\infty \left(\frac{3x^2+x-1}{2x^3+1} - \frac{kx+2}{2x^2+5} \right) dx$$

28. Identify which of the following improper integrals is **divergent**: (a) $\int_1^\infty \frac{e^{\sin x^3}}{x^2} dx$ (b) $\int_0^\infty \frac{1}{\sqrt{x^6+1}} dx$ (c) $\int_1^\infty \frac{\ln x}{e^{2x}} dx$ (d) $\int_2^\infty \frac{1}{(\ln x)^2} dx$

29. Identify which of the following improper integrals is **convergent**: (i) $\int_1^\infty \frac{\sqrt{x}}{1+x} dx$ (ii) $\int_1^\infty \frac{\ln x^2}{4+x^2} dx$ (iii) $\int_1^\infty \frac{2^x}{x+2^x} dx$ (iv) $\int_1^\infty \frac{1}{1+x \ln x} dx$ (v) $\int_1^\infty \frac{1}{x \ln \sqrt{x}} dx$

Part E: Exact Sums & Limits (Riemann Sums & Telescoping Series)

30. Evaluate the limit of the sequence of sums:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{i}{n^2 + i^2}$$

31. Find the exact sum of the series:

$$\sum_{n=2}^{\infty} \ln \left(1 - \frac{1}{n^2} \right)$$

32. Evaluate the limits of the following sequences:

(a) $\lim_{n \rightarrow \infty} \cos \frac{n\pi}{2n+1}$ (b) $\lim_{n \rightarrow \infty} \frac{\ln n}{\ln(3n)}$

33. Find the limit L of the recursive sequence $a_{n+1} = 6 - \frac{4}{a_n}$ with initial value $a_1 = 2$.

34. Identify which of the following infinite sequences converge to a finite limit as $n \rightarrow \infty$:

(i) $e^{\pi/n}$ (ii) $\frac{n^n}{n!}$ (iii) $\sqrt{n^2 + 2n} - n$ (iv) $\frac{(\ln n)^2}{n}$ (v) $\frac{(2n)!}{(n!)^2}$

35. Evaluate the exact sums of the following series:

(a) $\sum_{n=1}^{\infty} \frac{4(3^{n+1}) + 2^{n+1}}{6^n}$ (b) $\sum_{n=1}^{\infty} \frac{n}{2^{n-1}}$

36. Evaluate the sum of the telescoping series:

$$\sum_{n=1}^{\infty} \left(\left(\frac{1}{2} \right)^{\frac{1}{n+1}} - \left(\frac{1}{2} \right)^{\frac{1}{n}} \right)$$

37. Use the Alternating Series Estimation Theorem to determine the minimum number of terms required to estimate the sum of the series $S = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ to a specified accuracy.

Part F: Applications of Improper Integrals (Volume & Area)

38. Calculate the volume of the solid of revolution defined by the improper integral:

$$V = \int_0^{\infty} 2\pi x \cdot x^2 e^{-x^2} dx$$

39. Determine whether the volume of the solid defined by the improper integral $\int_0^{\infty} \pi x^4 e^{-2x^2} dx$ is a finite numerical value.

40. Find the area of the region bounded below the graph of the function $y = \frac{4x}{(x^2+1)^3}$ over the interval $0 \leq x < \infty$.

41. The region bounded below the curve $y = 9e^{-3x}$ for $0 \leq x < \infty$ is rotated about the y-axis. Calculate the volume of the generated solid.

42. The region below the curve $y = 2xe^{-x^3/6}$ over the interval $[0, \infty)$ is rotated about the x-axis. Calculate the volume of this solid.

- 43.** Given the volume of a solid of revolution expressed as $V = \int_0^\infty \frac{64\pi}{e^{2x}+4} dx$, evaluate its exact value. Determine if the integral $\int_0^\infty \frac{16\pi x}{\sqrt{e^{2x}+4}} dx$ converges to a finite sum.

Part G: Power Series & Taylor/Maclaurin Series

44. Consider the power series defined by $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \frac{1}{4^{2n}} (x-2)^{2n}$:
- (a) Find the interval of absolute convergence.
 - (b) Determine the convergence behavior at the boundary points $x = -2$ and $x = 6$.
 - (c) Based on this power series, evaluate the exact value of the definite integral $\int_2^3 f(x) dx$.
45. For the function defined by the power series $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n} (x-2)^n$:
- (a) Find the largest open interval where the power series converges absolutely (interior of the interval of convergence).
 - (b) Check the convergence at the two boundary points of the interval.
 - (c) Define $H(x) = (x-2)^2 f(x)$. Calculate the value of the derivative $H'(4)$.
46. Find the coefficient of the x^5 term in the Maclaurin series (Taylor series centered at 0) expansion of $f(x) = \ln(2+x^2)^2$.
47. Find the coefficient of the x^4 term in the Maclaurin series expansion of the composite function $f(x) = (1+8x^2)^{3/2}$.